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Fan assembly

Abstract

A battery powered box-type fan assembly includes a fan housing that at least partially defines an airflow chamber, a plurality of fan blades positioned within the airflow chamber, a motor operable to rotate the plurality of fan blades, a battery receptacle positioned on the fan housing, the battery receptacle configured to receive a removable battery pack to power the motor, and a base coupled to the fan housing and configured to be placed on a support surface. The base elevates the fan housing above the support surface and is operable to support the fan housing in multiple configurations. The base also includes a stand pivotable relative to the fan housing between a collapsed position, in which the stand is flipped up toward the fan housing, and a deployed position, in which the stand extends away from the fan housing and is configured to contact the support surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/160,305, filed Mar. 12, 2021, and to U.S. Provisional Patent Application No. 63/294,935, filed Dec. 30, 2021, the entire contents of which are incorporated by reference herein.

FIELD

(1) The present disclosure relates to cooling systems and, more particularly, to portable fan assemblies configured to be used in a workspace.

SUMMARY

(2) In one embodiment, the invention provides a battery powered box-type fan assembly including a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a battery receptacle positioned on the fan housing, the battery receptacle configured to receive a removable battery pack to power the motor; and a base coupled to the fan housing and configured to be placed on a support surface, the base elevating the fan housing above the support surface and operable to support the fan housing in multiple configurations. The base includes a stand pivotable relative to the fan housing between a collapsed position, in which the stand is flipped up toward the fan housing, and a deployed position, in which the stand extends away from the fan housing and is configured to contact the support surface.

(3) In another embodiment, the invention provides a battery powered box-type fan assembly including a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a battery receptacle positioned on the fan housing, the battery receptacle configured to receive a removable battery pack to power the motor; and a base coupled to the fan housing and configured to be placed on a support surface. The base includes a body elevating the fan housing above the support surface and operable to support the fan housing in multiple configurations, a stand movable relative to the body, and a latch coupled to the body and the stand, the latch releasably securing a position of the stand relative to the body, the latch being manually operable to allow movement of the stand relative to the body.

(4) In another embodiment, the invention provides a battery powered box-type fan assembly including a fan housing having a first face, a second face opposite the first face, and a perimeter extending between the first and second faces, the first face, the second face, and the perimeter defining a volume therebetween, the fan housing at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a base pivotally coupled to the fan housing and configured to be placed on a support surface, the base elevating the fan housing above the support surface, wherein the fan housing is pivotable relative to the base about a pivot axis that is parallel to the support surface; a battery receptacle positioned on the fan housing and electrically coupled to the motor; and a battery pack removably coupled to the battery receptacle, the battery pack positioned within the volume of the fan housing and beneath the pivot axis when coupled to the battery receptacle.

(5) In still another embodiment, the invention provides a battery powered fan assembly including a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a battery

receptacle configured to receive a removably battery pack to power the motor; and a base assembly configured to be placed on a support surface, the base assembly including a first frame segment coupled to the fan housing, a second frame segment pivotally coupled to the first frame segment, and a third frame segment pivotally coupled to the second frame segment. The base assembly is configured to be collapsed into a storage configuration in which each of the first, second, and third frame segments are folded onto each other, the base assembly is configured to be expanded to a first deployed configuration in which the fan housing is elevated above the support surface by a first height, and the base assembly is configured to be expanded to a second deployed configuration in which the fan housing is elevated above the support surface by a second height that is different than the first height.

(6) In another embodiment, the invention provides a battery powered fan assembly including a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the fan blades; a battery receptacle configured to receive a removable battery pack to power the motor; and a base assembly configured to be placed on a support surface, the base assembly including a first frame segment coupled to the fan housing, a second frame segment having a first end and a second end opposite the first end, the first end pivotally coupled to the first frame segment, a third frame segment pivotally coupled to the second end of the second frame segment, and a self-locking latch positioned between the second frame segment and the third frame segment, the self-locking latch biased into a locked position to inhibit relative movement between the second frame segment and the third frame segment.

(7) In yet another embodiment, the invention provides a battery powered fan assembly including a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the fan blades; a battery receptacle configured to receive a removable battery pack to power the motor; and a base assembly configured to be placed on a support surface, the base assembly including a first frame segment coupled to the fan housing, the first frame segment having a first reduced size gripping portion, a second frame segment pivotally coupled to the first frame segment, the second frame segment having a second reduced size gripping portion, and a third frame segment pivotally coupled to the second frame segment. The base assembly is configured to be collapsed into a storage configuration in which each of the first, second, and third frame segments are folded onto each other, and the base assembly is configured to be expanded to a deployed configuration in which the first, second, and third frame segments are unfolded from one another.

(8) In still another embodiment, the invention provides a fan assembly including a base including walls that at least partially define a storage area therebetween; a fan housing that at least partially defines an airflow chamber, the fan housing moveably coupled to the base between a stored position, in which the fan housing is positioned within the storage area, and a deployed position, in which the fan housing is at least partially positioned outside of the storage area; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a battery receptacle configured to receive a removable battery pack to power the motor; a mating interface formed on at least one of the base and the fan housing, the mating interface configured to interact with a complementary mating interface formed on a stackable element of the stackable storage system; and a latch mounted to the base, the latch configured to selectively inhibit relative movement between the fan assembly and the stackable element while the mating interface of the fan assembly interacts with the complementary mating interface of the stackable element.

(9) In another embodiment, the invention provides a fan assembly including a base including a first wall having a first mating interface, the first mating interface configured to interact with another stackable element of the stackable storage system, a second wall opposite the first wall and having a second mating interface, the second mating interface configured to interact with another stackable element of the stackable storage system, and a side wall extending between the first wall and the

second wall, the side wall defining an opening to a storage area in the base. The fan assembly further including a fan housing that at least partially defines an airflow chamber, the fan housing being moveably coupled to base between a stored position, in which the fan housing is positioned within the storage area, and a deployed position, in which the fan housing is at least partially extended out of the storage area through the opening; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; and a battery receptacle configured to receive a removable battery pack to power the motor.

(10) In still yet another embodiment, the invention provides a fan assembly including a base including walls that at least partially define a storage area therebetween; a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the fan blades; a battery receptacle configured to receive a removable battery pack to power the motor; and a coupler including a first end pivotally connected to the base about a first axis, and a second end rotatably connected to the fan housing about a second axis that is transverse relative to the first axis. The fan housing is configured to be selectively pivoted relative to the base about the first axis to move the fan housing into and out of the storage area, and the fan housing is configured to be selectively rotated relative to the base about the second axis to change a direction of airflow generated by the plurality of fan blades.

(11) Other aspects of the invention will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a portable fan assembly, according to one embodiment.

(2) FIG. 2 is a perspective view of the portable fan assembly of FIG. 1, illustrating a fan head in a rotated position.

(3) FIG. 3 is a perspective view of the portable fan assembly of FIG. 1 illustrating a fan head in a rotated position of about 180 degrees.

(4) FIG. 4 is a perspective view of a portable cooling system including one or more fan assemblies, according to another embodiment.

(5) FIG. 5 is a side perspective view of a single fan assembly of FIG. 4 supported on different support faces.

(6) FIG. 6 is a side perspective view of the portable cooling system of FIG. 4, illustrating multiple fan assemblies in a stacked configuration.

(7) FIG. 7 is a perspective view of a portable fan assembly, according to another embodiment.

(8) FIG. 8 is a side view of the portable fan assembly of FIG. 7 in multiple configurations.

(9) FIG. 9 is a perspective view of the portable fan assembly of FIG. 7 in a standing configuration.

(10) FIG. 10 includes perspective views of a portable fan assembly, according to another embodiment, illustrating the fan assembly in a standing configuration and a hanging configuration.

(11) FIG. 11 includes perspective views of the portable fan assembly of FIG. 10, illustrating the fan assembly in a collapsed configuration.

(12) FIG. 12 is another perspective view of the portable fan assembly of FIG. 10 in the collapsed configuration.

(13) FIG. 13 includes perspective views of the portable fan assembly of FIG. 10, illustrating the fan assembly in the standing configuration and the hanging configuration.

(14) FIG. 14 includes perspective views of a portable fan assembly, according to another embodiment, illustrating the portable fan assembly in multiple extended positions.

(15) FIG. 15 is perspective view of the portable fan assembly of FIG. 14, illustrating the portable fan assembly in a collapsed position.

(16) FIG. 16 includes perspective views of a portable fan assembly, according to another embodiment, illustrating the portable fan assembly in multiple configurations.

(17) FIG. 17 includes perspective views of the portable fan assembly of FIG. 16, illustrating the portable fan assembly in example applications.

(18) FIG. 18A is a perspective view of the portable fan assembly of FIG. 16, illustrating an alternate base assembly useable with the portable fan assembly.

(19) FIG. 18B is a side perspective view of the portable fan assembly of FIG. 18A, illustrating the alternate base in a flipped-down configuration.

(20) FIG. 18C is a side perspective view of the portable fan assembly of FIG. 18A, illustrating the alternate base in a flipped-up configuration.

(21) FIG. 18D is a perspective view of FIG. 18C.

(22) FIG. 18E is a side perspective view of the portable fan assembly of FIG. 18A, illustrating a fan housing in a first orientation.

(23) FIG. 18F is a side perspective view of the portable fan assembly of FIG. 18A, illustrating the fan housing in a second orientation.

(24) FIG. 18G is a perspective view of the portable fan assembly of FIG. 18B, illustrating a filter.

(25) FIG. 19A is a perspective view of a portion of the portable fan assembly of FIG. 18B.

(26) FIG. 19B is an isolated perspective view of a latch portion of the portable fan assembly of FIG. 19A, illustrating a latch in a locked position.

(27) FIG. 19C is an isolated perspective view of the latch portion of the portable fan assembly of FIG. 19A, illustrating the latch in an unlocked position.

(28) FIG. 19D is a perspective view of the portable fan assembly of FIG. 19A, illustrating the base in an intermediate position.

(29) FIG. 19E is a perspective view of the portable fan assembly of FIG. 19A, illustrating the base locked in a flipped-down position.

(30) FIG. 20A is a perspective view of a portable fan assembly, according to another embodiment.

(31) FIG. 20B is a perspective view of a self-locking latch of the portable fan assembly of FIG. 20A.

(32) FIG. 20C is a side view of the portable fan assembly of FIG. 20A, illustrating the portable fan assembly in a collapsed configuration.

(33) FIG. 20D includes side perspective views of the portable fan assembly of FIG. 20A, illustrating the portable fan assembly in the collapsed configuration and in a deployed configuration.

(34) FIG. 20E includes perspective views of the portable fan assembly of FIG. 20A, illustrating the portable fan assembly in example deployed configurations.

(35) FIG. 21A is a perspective view of a portable fan assembly, according to another embodiment.

(36) FIG. 21B is a perspective view of a portable fan assembly of FIG. 21A, illustrating the portable fan assembly being carried in a collapsed configuration.

(37) FIG. 21C is a perspective view of the portable fan assembly of FIG. 21A, illustrating the portable fan assembly being carried in the collapsed configuration, indicating a direction of rotation for deployment of the portable fan assembly.

(38) FIG. 21D is a perspective view of the portable fan assembly of FIG. 21A, illustrating the portable fan assembly in the collapsed configuration, indicating a direction of rotation for deployment of a base of the portable fan assembly.

(39) FIG. 21E is a perspective view of the portable fan assembly of FIG. 21A, illustrating the portable fan assembly with the base deployed, indicating a direction of deployment for a fan housing.

(40) FIG. 21F is a perspective view of the portable fan assembly of FIG. 21A, illustrating the portable fan assembly in a deployed configuration.

(41) FIG. 21G is a perspective view of the portable fan assembly of FIG. 21A, illustrating a joint of

the portable fan assembly.

(42) FIG. 22A is a perspective view of a fan assembly, according to another embodiment.

(43) FIG. 22B is a perspective view of the fan assembly of FIG. 22A in one deployed configuration.

(44) FIG. 22C is a perspective view of the fan assembly of FIG. 22A in another deployed configuration.

(45) FIG. 22D is a perspective view of the fan assembly of FIG. 22A in still another deployed configuration.

(46) FIG. 22E is a perspective view of the fan assembly of FIG. 22A in a collapsed configuration.

(47) FIG. 22F is a perspective view of the fan assembly of FIG. 22A, illustrating the fan assembly stacked on top of another stacking element.

(48) FIG. 23A is a perspective view of a fan assembly, according to another embodiment.

(49) FIG. 23B is another perspective view of the fan assembly of FIG. 23A.

(50) FIG. 23C is a perspective view of the fan assembly of FIG. 23A, illustrating the fan assembly stacked on top of another stacking element.

(51) FIG. 23D includes perspective views illustrating the fan assembly of FIG. 23A in a collapsed configuration and multiple deployed configurations.

(52) FIG. 24 is a perspective view of a fan assembly, according to another embodiment.

(53) FIG. 25 is a perspective view of the fan assembly of FIG. 24, illustrating an open storage compartment on the fan assembly.

(54) Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. Use of “including” and “comprising” and variations thereof as used herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Use of “consisting of” and variations thereof as used herein is meant to encompass only the items listed thereafter and equivalents thereof. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings.

DETAILED DESCRIPTION

(55) With brief reference to FIGS. 1-24, each embodiment may be powered by a battery pack 1, also referred to as a removable battery pack. The battery pack 1 may be a power tool battery pack generally used to power a power tool, such as an electric drill, an electric saw, and the like (e.g., an 18 volt rechargeable battery pack, or an M18 REDLITHIUM battery pack sold by Milwaukee Electric Tool Corporation). The battery pack 1 may include lithium ion (Li-ion) cells. In alternate embodiments, the battery packs may be of a different chemistry (e.g., nickel-cadmium (NiCa or NiCad), nickel-hydride, and the like). The battery pack 1 may be an 18 volt battery pack, a 4 volt battery pack, a 28 volt battery pack, a 40 volt battery pack, or a battery pack of any other voltage such that the capacity of the battery pack may vary. The battery pack may further include an indicator to display the current state of charge of the battery pack and/or other characteristics of the battery pack.

(56) FIGS. 1-3 illustrate a portable cooling system or fan assembly 10 according to one embodiment. The fan assembly 10 is configured to provide airflow to an area such as a workspace, a workshop, a jobsite, or the like. The fan assembly 10 may include a main body 14 that supports a user interface 18 and a handle 22 configured to accommodate carrying the fan assembly 10. The user interface 18 may be a button, a switch, a toggle, or another selector that allows a user to interact and thus control an operation of the fan assembly 10. In the illustrated embodiment, the fan

assembly **10** has an overall rectangular shape, although other polygonal shapes are considered. In some embodiments, the fan assembly **10** may be generally curved and/or oblong.

(57) As illustrated in FIG. 1, the main body **14** is generally cuboid and includes at least four support faces **26A**, **26B**, **26C**, **26D** upon which the fan assembly **10** may be supported when, for example, the fan assembly **10** is set on a surface such as a floor surface, table top, or the like. As illustrated in FIGS. 2 and 3, the main body **14** may alternatively have a narrowed or thinned width, thickness, or the like. The fan assembly **10** further includes a first fan head **30A** and a second fan head **30B**. The fan heads **30A**, **30B** may each be rotatably coupled to opposing faces **26E**, **26F** of the main body **14** and rotatable relative the main body **14** about an axis A-A. The first fan head **30A** may support a first fan **34A**, such as a blade(s), impeller, or the like rotatably supported therein. The second fan head **30B** may support a second fan **34B**, such as a blade(s), impeller, or the like rotatably supported therein. The first fan **34A** and the second fan **34B** may be rotated by a battery powered motor or otherwise powered (e.g., wired connection) motor to generate airflow in a desired direction.

(58) Each fan head **30A**, **30B** may also be polygonal shaped. In the illustrated embodiment, the main body **14** and fan heads **30A**, **30B** are similarly shaped and may have similar dimensions. Namely, the support faces **26A**, **26B**, **26C**, **26D** of the main body **14** may selectively align with faces **32** of the fan heads **30A**, **30B** to give the fan assembly **10** a generally rectangular shape. In the illustrated embodiment, each fan head **30A**, **30B** is generally cuboid.

(59) As best shown in FIG. 2, the fan heads **30A**, **30B** are rotatable about the axis A-A relative to one another, as well as relative the main body **14**. Each fan head **30A**, **30B** may further support a shroud **38** and a grille **42**. In one example construction of the fan assembly **10**, the blades of the first fan **34A** and the second fan **34B** are rotated to generate an airflow **46** generally entering the respective fan heads **30A**, **30B** through a rear portion and along the shrouds **38** and exiting through the grilles **42**.

(60) As further shown in FIG. 2, a direction of airflow **46** may be selectively changed by rotating one of the fan heads **30A**, **30B** or by rotating the fan assembly **10**. In some scenarios, as shown in FIG. 3, the direction of the airflow **46** generated by each fan head **30A**, **30B** may be relatively offset by about 180 degrees. In other scenarios, the direction of the airflow **46** generated by each fan head **30A**, **30B** may be relatively offset by about another about between 0 degrees and 180 degrees, including 0 degrees.

(61) As even further shown in FIG. 2, the fan assembly **10** may be supported on any face **32** of the fan heads **30A**, **30B** including a face **32** supporting the grille **42**, a face **32** intersecting the axis A-A, or another face **32**. Each fan head **30A**, **30B** may further support one or more auxiliary handles **50**. The auxiliary handles **50**, similar to the handle **22**, may be configured to accommodate carrying the fan assembly **10**. The user interface **18** may be operable to select a rotational speed and/or rotational direction of the blades of the fans **34A**, **34B**. In the illustrated embodiment, the user interface **18** is operable to selectively control both of the fans **34A**, **34B** together, or each of the fans **34A**, **34B** independently.

(62) FIGS. 4-6 illustrate a portable cooling system **100** including one or more fan assemblies **104** according to another embodiment. The fan assembly **104** supports a fan **110** rotatably mounted therein. The fan **110** is configured to generate an airflow **114** through a fan body or shroud **118** and out a grille **122** positioned on a face of the fan assembly **104**. As illustrated in FIG. 5, the fan assembly **104** may be formed of opposing first and second support faces **124A**, **124B** and opposing third and fourth support faces **128A**, **128B**. The support faces **124A** . . . **128B** are generally not faces that support the grille **122**, but the fan assembly **104** could still be supported by one of such faces.

(63) The first and second support faces **124A**, **124B** may be angled relative to the third and fourth support faces **128A**, **128B** such that a user may select an angle of airflow **114** direction by altering which support face **124A** . . . **128B** the fan assembly **104** is supported on. For example, when the

fan assembly **104** is supported on the third or fourth support face **128A**, **128B**, airflow **114** may be generally parallel to a surface **132** upon which the fan assembly **104** is supported. When the fan assembly **104** is supported on the first support faces **124A** and/or the second support faces **124B**, airflow **114** may be generally offset relative the surface **132** upon which the fan assembly **104** is supported. In one embodiment, the airflow **114** is offset relative the surface **132** by about 4.5 degrees when the fan assembly **104** is supported on the first support faces **124A** or the second support faces **124B**.

(64) The portable cooling system **100** may include a plurality of fan assemblies **104** that are stackable relative one another at an interface **142**. The interface **142** may include complimentary mating features **146** formed on opposing faces of adjacently stacked fan assemblies **104**. The mating features **146** allow the fan assemblies **104** to be stacked and retained in a stable stacked configuration.

(65) Each fan assembly **104** may further include a filter slot for replaceably receiving a filter, such as a standard 20 inch furnace filter. The fan assemblies **104** may receive power from the battery pack **1** and supply an AC or DC output through a ground fault circuit interrupting (GFCI) 120V outlet **152**. Each fan assembly **104** may even further include a variable speed user interface **156**. In some embodiments, each fan assembly **104** may have an overall square dimension of approximately twenty inches.

(66) FIGS. 7-9 illustrate a fan assembly **200**, such as a portable fan assembly, according to another embodiment. The fan assembly **200** may include a shroud or housing **204**, a grille **208**, and a fan **212** rotatably supported within the housing **204**. Legs **216** may be rotatably coupled to the fan assembly **200** to support the fan assembly **200** in multiple configurations. In the illustrated embodiment, the legs **216** are rotatably coupled to the housing **204**. The fan **212** may be rotated by a battery powered motor or otherwise powered (e.g., wired connection) motor to generate airflow in a desired direction.

(67) As shown in FIG. 8, the legs **216** allow the fan assembly **200** to be operated and/or stored in multiple configurations including an elevated and tilted configuration **220**, an elevated and level configuration **224**, a first low height tilted configuration **228** in which the fan assembly **200** is supported on a surface by a portion of either leg **216** and the housing **204**, a second low height tilted or level configuration **232** in which the fan assembly **200** is supported on the surface solely by either leg **216**, and a collapsed configuration **240**. A height of the fan assembly **200** in any elevated configuration (e.g., elevated and level configuration **224**) may be similar to that of a user (FIG. 9). The legs **216** further allow for general universal adjustment of the fan assembly **200** to thereby direct airflow from the fan **212** in a desired direction.

(68) Rotation of the legs **216** relative the housing **204** may be locked in a desired position via leg locks **244**. The fan assembly **200** may further include handles **248** for accommodating carrying the fan assembly **200**, one or more user interfaces **252** for controlling an operational output of the fan assembly **200**, one or more battery packs **1**, a filter (e.g., standard 20×20×1 inch air filter), and one or more rotatable ground engaging elements **256** (e.g., wheels, casters, etc.). One or both of the handles **248** may be a flexible material. One or both of the handles **248** may also be rigid and selectively wrapped with a cord, tape, or the like.

(69) FIGS. 10-13 illustrate a fan assembly **300**, such as a portable fan assembly, according to another embodiment. The fan assembly **300** may include a shroud or housing **304**, a grille **308**, and a fan **312** (e.g., blades) rotatably supported within the housing **304**. One or more support bars **316** may be rotatably coupled to the fan assembly **300** to support the fan assembly **300** in multiple configurations. In the illustrated embodiment, the support bar **316** is rotatably coupled to the housing **304**. The fan **312** may be rotated by a battery powered motor or otherwise powered (e.g., wired connection) motor to generate airflow in a desired direction.

(70) As shown in FIGS. 10 and 13, the support bar **316** allows the fan assembly **300** to be operated and/or stored in multiple configurations including an standing configuration **320** in which the fan

assembly **300** is supported on a surface by a portion of the support bar **316** and the housing **304**, a collapsed configuration **340** (FIGS. **11** and **12**), and a hanging configuration **344** in which the fan assembly **300** may be hung or otherwise supported on a structure by links or hooks **346** formed on the support bar **316** and extending therefrom. In one example, the hooks **346** may support the fan assembly **300** in the hanging configuration in which the hooks **346** at least partially wrap around a beam or plank such as a wooden stud. The hooks **346** may also be rotatably coupled to the support bar **316** such that the hooks **346** may be rotated relative the support bar **316** between collapsed (FIGS. **11** and **12**) and extended positions.

(71) The fan assembly **300** may further include handles **348** for accommodating carrying the fan assembly **300**, one or more user interfaces **352** for controlling an operational output of the fan assembly **300**, and one or more battery packs. In one example construction, the fan assembly **300** may have overall dimensions of 16 inches tall (H), 13.5 inches wide (W), and 3.5 inches deep (D).

(72) FIGS. **14** and **15** illustrate a fan assembly **400**, such as a portable fan assembly, according to another embodiment. The fan assembly **400** may include a shroud or housing **404**, a grille **408**, and a fan (e.g., blades) **412** rotatably supported within the housing **404**. The housing **404** may be adjustably supported by a base **416** having a support beam **418**. The base **416** may include telescoping members **420** that accommodate raising and lowering of the housing **404** relative the support beam **418**. In the illustrated embodiment, the support beam **418** may be rotatably or otherwise adjustably attached to the telescoping members **420** between a collapsed position (FIG. **15**) and an extended position in which the support beam **418** accommodates standing of the fan assembly **400** on a surface. The fan **412** may be rotated by a battery powered motor or otherwise powered (e.g., wired connection) motor to generate airflow in a desired direction.

(73) In one example construction, the base **416** accommodates height adjustment of the housing **404** to different heights between about twenty inches tall and about fifty inches tall, including twenty inches and fifty inches. In another example construction, the base **416** accommodates height adjustment of the housing **404** to heights above a typical workspace fan such as higher than the height of an average user (e.g., 5 feet 8 inches to 7 feet).

(74) The base **416** may further include a pedal **424** that may be operated by a user to selectively lock and unlock the telescoping members **420** relative to the support beam **418**. The pedal **424** may be a foot pedal **424** that is configured to be pressed by a foot of a user. For example, the user may step on the pedal **424** to thereby release the telescoping members **420**. The user may then lift the housing **404** to a desired height while maintaining pressure on the pedal **424**. Once the user releases the pedal **424**, the telescoping members **420** may once again be constrained from movement.

(75) In some embodiments the telescoping members are lift actuated by a spring, hydraulic, plunger, or another type of actuator. In some embodiments, the pedal is biased to a locked position. The fan assembly **400** may further include handles **448** for accommodating carriage of the fan assembly **400**, one or more user interfaces **452** (e.g., knobs, buttons, switches, etc.) for controlling an operational output of the fan assembly **400**, and one or more battery packs **1** configured to power the motor.

(76) FIGS. **16-19E** illustrate a fan assembly **500**, such as a portable and battery powered box-type fan assembly, according to another embodiment. The fan assembly **500** includes a fan housing **504**, a grille **508**, a fan **512** (such as fan impeller, a blade, or a plurality of fan blades) rotatably supported within the fan housing **504**, and a base **516**. The base **516** is configured to be placed on a support surface (e.g., floor surface, table top, windowsill, etc.) and elevate the fan housing **504** above the support surface to not disturb dust or debris below. The fan housing **504** is rotatably coupled to the base **516** such that the base **516** is operable to support the fan housing **504** in multiple directions, orientations, and/or configurations.

(77) The fan **512** may be rotated by a motor **514** that can be powered by a DC power source (e.g., a power tool battery back), such as the removable battery pack **1**, or an AC power source (e.g., a wall outlet). The fan **512** is positioned within an airflow chamber **505** at least partially defined by the

fan housing **504**. In the illustrated embodiment, the fan **512** is rotated by the motor **514** to generate airflow through the airflow chamber **505**. The fan **512** may be rotated by the motor **514** about a rotation axis **506**.

(78) The fan assembly **500** further includes a battery receptacle **515** that can receive the removable battery pack **1**. In the illustrated embodiment, the battery receptacle **515** is positioned on the fan housing **504**. The fan housing **504** may further support an AC power inlet **517** that can provide charging power to the removable battery pack **1** that is received in the battery receptacle **515**. In some embodiments, the AC power inlet **517** is positioned on another part of the fan assembly **500**, such as the base **516**. In either position, the AC power inlet **517** is electrically coupled to the motor **514**, to the battery receptacle **515**, or to both, such that the motor **514** could be operated on power provided from the battery pack **1** and/or from the AC power inlet **517**. An operational output (e.g., speed) of the motor **514** may be controlled by a selector **513**, such as a rotational knob, a switch, a button, and/or the like.

(79) Referring to FIG. **16**, the base **516**, or frame, includes a stand **518** that supports stabilizing legs **520** to keep the fan housing **504** upright during use. The stabilizing legs **520** fold out or flip-down to provide extra stability in a deployed position, and fold in or flip-up for slender storage and transport in a collapsed position. In the deployed positions, the stabilizing legs **520** are configured to contact the support surface. As shown in FIG. **16**, the stabilizing legs **520** may be coupled to the fan housing **504** by hinges **522** such that the stabilizing legs **520** rotate 90 degrees between the deployed and collapsed positions. Brackets **524** may also lock the stabilizing legs **520** in place when deployed. In other embodiments, as illustrated in FIG. **17**, the brackets may be omitted. Additionally or alternatively, the hinges **522** may each include a one-way locking detent latch **526**, as shown in FIG. **19A** and described below. In the illustrated embodiment, a pair of stabilizing legs **520** are positioned on each side of the fan housing **504**. In other embodiments, stabilizing legs **520** are positioned on a single side of the fan housing **504**.

(80) As shown in FIGS. **16**, **18E**, and **18F**, the fan housing **504** is pivotal (e.g., rotatable) relative to the base **516** about a pivot axis **507** that is parallel to the support surface and perpendicular to the rotation axis **506**. The stand **518** may include a body **536** and one or more (e.g., two) arms **527** connected by the body **536**. In the illustrated embodiment, the body **536** is elongated along a width of the fan housing **510**. The arms **527** may extend upwardly from the body **536** and on diametrically opposite sides of the airflow chamber **505** to support the fan housing **504** through a plurality of directions. For example, the fan housing **504** may rotate to adjust an angle of the fan **512** to alter a direction of the airflow. In some embodiments, the fan housing **504** and/or base **516** may have hard stops **519** (FIGS. **18E** and **18F**) to inhibit over-rotation relative to each other. For example, the fan housing **504**/base **516** may have a first hard stop **519** at a 20 degree downward tilt (FIG. **18E**) and a second hard stop **519** at a 50 degree or 90 degree upward tilt (FIG. **18F**).

(81) The fan housing **504** may further include a handle **528**. The handle **528** may fold down into the fan housing **504**. As illustrated in FIG. **18G**, the fan housing **504** may also include retainers **530**, such as clips, magnets, tracks, and/or the like, located behind the fan **512**. The retainers **530** may allow a standard filter (e.g., a 20 inch×20 inch×1 inch filter) to be releasably held behind the fan **512**. In the illustrated embodiment, the filter may be placed adjacent an inlet of the airflow chamber **505**.

(82) As shown in FIG. **17**, the fan assembly **500** may include mounting features **532** to help mount the fan assembly **500** in a window opening. For example, the fan assembly **500** may include a bungee cord **533** or other suitable cord or cable. Additionally or alternatively, the fan assembly **500** may include a hook **534** or other suitable projection. While mounted in a window opening, a front pair of the stabilizing legs **520** and/or a rear pair of the stabilizing legs **520** may be flipped-up. For example, the rear pair of stabilizing legs **520** may be flipped up, and the fan assembly **500** may be at least partially supported by a frame of the window or another portion of the window. In one instance, the fan assembly **500** could be leaned against a screen or frame within an opening

window.

(83) Referring now to FIGS. **18A-19E**, the fan housing **504** may be supported by an alternate base **516a** that is similar to the base **516** illustrated in FIGS. **16** and **17**. The alternate base **516a** is coupled to the fan housing **504** and is placed on the support surface and includes a body **536** that elevates the fan housing **504** above the support surface. The alternate base **516a** and body **536** are also operable to support the fan housing **504** in multiple configurations, orientations, directions, etc. The alternate base **516a** further includes a stand **540** movable relative to the body **536** and the one-way locking detent latch **526**, which is coupled to the body **536** and to the stand **540**. The detent latch **526** releasably secures a position of the stand **540** relative to the body **536** and is operable (e.g., by a user) to allow movement of the stand **540** relative to the body **536**. The detent latch **526** is further positioned adjacent at least one of the hinges **522**.

(84) As specifically illustrated in FIGS. **18A-18G**, the fan housing **504** has a first face **546**, a second face **548** opposite the first face **546**, and a perimeter extending between the first face **546** and the second face **548**. In other words, the perimeter is made up of the plurality of faces **550** extending between the first face **546** and the second face **548**. In the illustrated embodiment, the perimeter is polygonal (e.g., rectangular, square, hexagonal, etc.) but could be circular in other embodiments. In the illustrated embodiment, a perimeter of the airflow chamber **505** is circular, but could be polygonal in other embodiments.

(85) The first face **546**, the second face **548**, and the plurality of faces **550** (e.g. the perimeter) together define a volume therebetween. In the illustrated embodiment, the battery receptacle **515** is positioned within the volume adjacent the second face **548** of the fan housing **504**. In the illustrated embodiment, the first face **546** includes the retainers **530** for the replaceable filter. Referring specifically to FIG. **18A**, the battery receptacle **515** is positioned adjacent a vertex of two side faces (e.g., faces **550**) and below both the rotation axis **506** and the pivot axis **507**. The position of the battery receptacle **515** on the fan housing **504** provides additional support to the fan assembly **500** while the battery pack **1** is received in the battery receptacle **515**. In some instances, the position of the battery receptacle **515** makes movement of the fan housing **504** easier for the user and resists the fan housing **504** from being flipped over.

(86) In the illustrated embodiment, as illustrated in FIGS. **19B** and **19C**, the detent latch **526** includes a plurality of ramps **542a** each terminated by a shoulder **544a**. The stand **540** includes complementary ramps **542b** and shoulders **544b** that are radially offset relative the ramps **542a** and the shoulders **544a** such that the detent latch **526** and the stand **540** cooperate with one another in a one-way ratchet manner. As such the detent latch **526** includes a ratchet mechanism that has a series of ramps **542a** and shoulders **544a** that contact the complementary series of ramps **542b** and shoulders **544b** formed on the stand **540** to permit ratcheting rotation of the stand **540** relative to the body **536**.

(87) As further illustrated in FIGS. **19B** and **19C**, the stand **540** may be rotatably coupled to the body **536** with the hinge **522**, which may be an elongated hinge that extends along the width of the fan housing **504** between the diametrically opposite sides of the fan housing **504**. The hinge **522** may also attach a single pair of stabilizing legs **520** together. The detent latch **526** may be biased by a biasing member **554** (e.g., spring) and moveable axially along a portion of the hinge **522**. The biasing member **554** may bias the detent latch **526** into a locked position (FIG. **19B**) in which the stand **540** is moveable relative the body **536** in a first direction or flip-down direction **556** (arrow in FIG. **19D**) but inhibited from flipping up by contact between the shoulders **544a** on the detent latch **526** and the shoulders **544b** on the stand **540**. As the stand **540** moves in the flip-down direction **556**, the ramps **542a**, **542b** engage and slide on one another. As the ramps **542a**, **542b** slide, the biasing member **554** is engaged and allows the detent latch **526** to slide until the **542a**, **542b** clear one another and the shoulders **544a**, **544b** slide into engagement with one another, in which the detent latch **526** will prevent movement of the stand **540** in the opposing flip-up direction.

(88) Once the stand **540** is flipped down so that the stabilizing legs **520** are parallel with the

support surface, as illustrated in FIGS. 18B and 19E, the detent latch 526 is actuated, such as by the hand or foot of a user, to separate the shoulders 544a, 544b from one another, and allow the stand 540 to be flipped up (e.g., from the deployed position toward the collapsed position). In some embodiments, the alternate base 516a includes a stand 540 on each side of the fan housing 504, and each stand 540 is moveable independently from the other. In such embodiments, the fan assembly 500 includes a detent latch 526 on each stand 540.

(89) FIGS. 20A-21G illustrate a fan assembly 600, such as a portable fan assembly, according to another embodiment. The fan assembly 600 includes a fan housing 604, a grille 608, a fan 612 rotatably supported within the fan housing 604, and a base assembly or base 616. The fan 612 may be or include an impeller, a blade, a plurality of blades, or the like, and may be rotated by motor 614 powered by a DC power source, such as the battery pack 1, or an AC power source (e.g., a wall outlet). The fan 612 may be supported by the fan housing 604 within an airflow chamber at least partially defined by the fan housing 604. FIGS. 20A-20E illustrate a base 616 useable with the fan housing 604, and FIGS. 21A-21F illustrated an alternate base 616a useable with the fan housing 604 or a slightly modified fan housing.

(90) The base 616 and the alternate base 616a are configured to be placed on a support surface (e.g., floor surface, table top, windowsill, etc.) and elevate the fan housing 604 above the support surface to not disturb dust or debris below. The fan housing 604 is rotatably coupled to the base 616 and the alternate base 616a such that the base 616 and the alternate base 616a are operable to support the fan housing 604 in multiple directions, orientations, and/or configurations. The fan 612 is positioned within an airflow chamber 605 at least partially defined by the fan housing 604. In the illustrated embodiment, the fan 612 is rotated by the motor 614 to generate airflow through the airflow chamber 605. The fan 612 may be rotated by the motor 614 about a rotation axis 606.

(91) The fan assembly 600 further includes a battery receptacle 615 that can receive the removable battery pack 1. In the illustrated embodiment of FIGS. 20A-20E, the battery receptacle 615 is positioned on the fan housing 604. In the illustrated embodiment of FIGS. 21A-21G, the battery receptacle 615 is positioned on the alternate base 616a. The fan housing 604 and/or the alternate base 616a may further support an AC power inlet 617 that can provide charging power to the removable battery pack 1 that is received in the battery receptacle 615. In some embodiments, the AC power inlet 617 is positioned on another part of the fan assembly 600, such as the base 616. In either position, the AC power inlet 617 is electrically coupled to the motor 614, to the battery receptacle 615, or to both, such that the motor 614 could be operated on power provided from the battery pack 1 and/or from the AC power inlet 617.

(92) The base 616, or frame, includes a first frame segment 618a having a first end 619 and an opposite second end 620, a second frame segment 618b having a first end 621 and a second end 622, and a third frame segment 618c having a first end 623 and a second end 624. Each of the illustrated frame segments 618a, 618b, 618c includes two parallel, spaced apart beams or members. In other embodiments, each frame segment 618a, 618b, 618c may only include a single beam or member. The first frame segment 618a can be pivotally coupled to the second frame segment 618b, and the second frame segment 618b may be pivotally coupled to the third frame segment 618c. The first frame segment 618a, the second frame segment 618b, and the third frame segment 618c may be expanded (e.g., unfolded) into a plurality of deployed configurations (FIGS. 20A and 20E) and collapsed (e.g., folded, folded down, etc.) into a compact form or storage configuration (FIG. 20C). The fan housing 604 may rotate or pivot relative to the base 616. For example, the fan housing 604 may rotate 360 degrees relative to the base 616. In some embodiments, rotation of the fan housing 604 may be limited to 320 degrees. In the illustrated embodiment, the fan housing 604 is coupled to the first frame segment 618a, but could be alternately coupled to other portions of the base 616.

(93) The base 616 may elevate the fan housing 604, for example, 15 inches, 32 inches, or 42 inches off the ground when in either of the deployed configurations. As illustrated in FIG. 20E, the base 616 is expandable into a first deployed configuration (left view of FIG. 20E) in which the first

frame segment **618a** is folded on the second frame segment **618b**, the second frame segment **618b** is unfolded from the third frame segment **618c**, and the fan housing **604** is elevated above the support surface by a first height between approximately ten inches and approximately twenty inches (e.g., 15 inches). As also illustrated in FIG. 20E, the base **616** is expandable into a second deployed configuration (right view of FIG. 20E) in which the first frame segment **618a** is unfolded from the second frame segment **618b**, the second frame segment **618b** is unfolded from the third frame segment **618c**, and the fan housing **604** is elevated above the support surface by a second height between approximately thirty inches and approximately fifty inches (e.g., 32 inches, 42 inches, etc.). In other deployed configurations, the first frame segment **618a**, the second frame segment **618b**, and the third frame segment **618c** may be expanded into different configurations to achieve different heights of the fan housing **604**.

(94) As best illustrated in FIGS. 20A and 20C, the first frame segment **618a** may include a first gripping portion **626** and the second frame segment **618b** may include a second gripping portion **630**. The first gripping portion **626** and the second gripping portion **630** may be reduced in size relative to the adjacent portions of the base **616** and may aid in the set-up and adjustment of the fan assembly **600**. The first gripping portion **626** and the second gripping portion **630** may include abrasive or textured surfaces. As shown in FIG. 20C, the first gripping portion **626** and the second gripping portion **630** may be adjacent one another while the base **616** is collapsed into the storage configuration to provide a relief space for a user to grasp when unfolding or deploying the base **616**. For example, when the first gripping portion **626** and the second gripping portion **630** are adjacent to one another, a user may fit their fingers into the relief space provided to grasp the base **616**.

(95) Referring now to FIG. 20D, the base **616** further includes a first joint **632** connecting the fan housing **604** and the first frame segment **618a**, a second joint **634** connecting the first frame segment **618a** and the second frame segment **618b**, and a third joint **636** connecting the second frame segment **618b** to the third frame segment **618c**. In the illustrated embodiment, the first joint **632** is positioned between the first end **619** and the second end **620** of the first frame segment **618a**, the second joint **634** is positioned at the first end **623** of the second frame segment **618b**, and the third joint **636** is positioned at the second end **624** of the second frame segment **618b**. In some embodiments, the joints are positioned on other parts of the frame segments.

(96) With continued reference to FIG. 20D, the second frame segment **618b** is pivotal toward the storage configuration relative to the third frame segment **618c** in a first direction **640**, and the first frame segment **618a** is pivotal toward the storage configuration relative to the second frame segment **618b** in a second direction **644**. In the illustrated embodiment, the first direction **640** and the second direction **644** are opposite to one another, but could be generally similar to one another in other embodiments.

(97) Referring now to FIGS. 20A and 20B, the fan assembly **600** further includes a latch or self-locking latch **648** positioned between the second frame segment **618b** and the third frame segment **618c**. In some embodiments, the third joint **636** includes the self-locking latch **648**. In the illustrated embodiment, the self-locking latch **648** is biased into a locked position (left view of FIG. 20B) in which the self-locking latch **648** inhibits relative movement between the second frame segment **618b** and the third frame segment **618c**. The during deployment of the fan assembly **600**, the second frame segment **618b** is moveable relative to the third frame segment **618c** without the user needing to operate the self-locking latch **648**. The self-locking latch **648** has a ramp **650** and a shoulder **652** that are engaged by a ramp **654** and a shoulder **656** on the first end **623** of the second frame segment **618b**. The ramp **654** of the second frame segment **618b** displaces the ramp **650** of the self-locking latch **648** as the second frame segment **618b** is pivoted into the deployed condition.

(98) Once the ramp **654** of the second frame segment **618b** clears the ramp **650** of the self-locking latch **648**, the shoulder **652** of the self-locking latch **648** bears against the shoulder **656** of the second frame segment **618b** to prevent relative pivoting between the two. While locked, the self-

locking latch **648** may be operated (e.g., by the hand or foot of a user) into an unlocked or unlatched positioned (right view of FIG. **20B**) to allow relative pivoting between the second frame segment **618b** and the third frame segment **618c**. The fan assembly **600** may even further include a hinge or friction hinge **670** positioned between the first frame segment **618a** and the second frame segment **618b**. The friction hinge **670** is configured to inhibit the first frame segment **618a** and the second frame segment **618b** from swinging (e.g., pivoting) freely from one another. Stated another way, the friction hinge **670** internal pin or slide mechanisms to resist movement withing the second joint **634**. The friction hinge **670** may also be positioned in the first joint **632** and/or the third joint **636**. In general, the self-locking latch **648** or latches and the friction hinge **670** or hinges may keep the base **616** upright and locked while in use.

(99) As illustrate in FIG. **21A**, the fan assembly **600** may even further include a handle or carrying handle **664** supported on at least one of the fan housing **604** and the alternate base **616a**. In the illustrated embodiment, the carrying handle **664** is configured to accommodate carrying of the fan assembly **600** in any of the storage configuration, the first deployed configuration, and the second deployed configuration. With further reference to FIG. **21A**, the first frame segment **618a**, the second frame segment **618b**, and the third frame segment **618c** of the alternate base **616a** may be telescopically connected to one another such that the second joint **634** and the third joint **636** are sliding joints rather than pivoting joints. In the illustrated embodiment, pivoting or sliding joints may include friction hinges and/or self-locking latched.

(100) In some embodiments, as illustrated in FIG. **21G**, the alternate base **616a** may include a slider-pin latch **666** within the second joint **634**. The slider-pin latch **666** includes a spring assembly **671**, a retainer **672**, and a button **674** biased outwardly by the spring assembly **671**. The slider-pin latch **666** is positioned within one of the first frame segment **618a** and the second frame segment **618b**, and the button **674** is positioned to protrude outwardly from the second joint **634** to prevent sliding between the first frame segment **618a** and the second frame segment **618b**. The button **674** may be pressed by a user to rotate the retainer **672** out of preventing engagement between the first frame segment **618a** and the second frame segment **618b** and to allow relative sliding therebetween.

(101) With reference back to FIGS. **21B-21F**, the fan assembly **600** may be carried by a user via the carrying handle **664** (FIG. **21B**) while in the collapsed position. The user may rotate the fan assembly **600** upward (FIG. **21C**) and set on the support surface (FIG. **21D**). The user may then place the base **616** or the alternate base **616a** while placing their foot on a portion of the base **616** or the alternate base **616a** (FIG. **21E**). The user may then grasp portions of the fan assembly **600**, such as the first gripping portion **626**, the second gripping portion **630**, the first joint **632**, etc. and lift the fan housing **604** relative to the base **616** or the alternate base **616a** to move the fan assembly **600** from the collapsed configuration to the deployed configuration (FIG. **21F**). For example, the user may press the button **674** or buttons to allow movement between the first frame segment **618a** and the second frame segment **618b** and move the fan assembly **600** toward the deployed configuration. In some embodiments, the fan assembly **600** includes flip-up/flip-down legs **665** that are selectively moveably relative the base **616** (e.g., the third frame segment **618c**) or alternate base **616a** to provide more stability to the fan assembly **600**.

(102) FIGS. **22A-23D** illustrate a fan assembly **700**, such as a portable fan assembly, according to another embodiment. The fan assembly **700** includes a fan housing **704**, a grille **708**, a fan **712** such as fan impeller, a blade, or a plurality of fan blades, rotatably supported within the fan housing **704**, and a base **716**. The fan **712** may be powered by a DC power source (e.g., a power tool battery back) or an AC power source (e.g., a wall outlet). The fan assembly **700** also includes an AC pass through.

(103) The fan **712** may be rotated by a motor **714** that can be powered by a DC power source (e.g., a power tool battery back), such as the removable battery pack **1**, or an AC power source (e.g., a wall outlet). The fan **712** is positioned within an airflow chamber **705** at least partially defined by

the fan housing **704**. In the illustrated embodiment, the fan **712** is rotated by the motor **714** to generate airflow through the airflow chamber **705**. The fan **712** may be rotated by the motor **714** about a rotation axis **706**.

(104) The fan assembly **700** further includes a battery receptacle **715** that can receive the removable battery pack **1**. In the illustrated embodiment, the battery receptacle **715** is positioned on the fan housing **704**. The base **716** may further support an AC power inlet **717** that can provide charging power to the removable battery pack **1** that is received in the battery receptacle **715**. In some embodiments, the AC power inlet **717** is positioned on another part of the fan assembly **700**, such as the fan housing **704**. In either position, the AC power inlet **717** is electrically coupled to the motor **714**, to the battery receptacle **715**, or to both, such that the motor **714** could be operated on power provided from the battery pack **1** and/or from motor **714**. The fan **712** is positioned within an airflow chamber **705** at least partially defined by the fan housing **704**. In the illustrated embodiment, the fan **712** is rotated by the motor **714** to generate airflow through the airflow chamber **705**. The fan **712** may be rotated by the motor **714** about a rotation axis **706**.

(105) The base **716** may include walls **718** that define a storage area **720** or storage areas. For example, the base **716** may be a toolbox. The base **716** also includes a mating interface **724** configured to interact with other storage units, such as the Milwaukee Tool PACKOUT Modular Storage System. The storage area **720** may also be a space defined in the base **716** for receiving the fan housing **704**. As illustrated in FIGS. 22B-22E and 23D, the fan housing **704** is moveably coupled to the base **716** between a stored position (FIGS. 22E and 23A), in which the fan housing **704** is positioned within the storage area **720**, and a deployed position (FIGS. 22B-22D, 22F, and 23C), in which the fan housing **704** is at least partially positioned outside of the storage area **720**. In the illustrated embodiment, the fan housing **704** may pivot relative to the base **716** between the stored and deployed positions.

(106) In some embodiments, as illustrated in FIG. 22A-22F, 24, and 25, the mating interface **724** is formed on the fan housing **704**, such as on the grille **708** at least partially overlaying the airflow chamber **705**, and the base **716**. The fan assembly **700** further includes a latch **728** moveably mounted to the base **716** and configured to selectively inhibit relative movement between the fan assembly **700** and another stackable element **732**, such as a toolbox, rolling cart, container, and/or the like while the mating interface **724** of the fan assembly **700** interacts with a complementary mating interface of the stackable element **732**. When the mating interface **724** is formed on the grille **708**, the fan housing **704** is retained in the storage area **720** (e.g., in the stored position) while the stackable element **732** is stacked on top of the fan assembly **700**.

(107) As illustrated in FIGS. 23A-23D, the mating interface **724** is not formed on the grille **708**, but rather formed on the base **716** such that the fan housing **704** is moveable out of the storage area **720** (e.g., to the deployed position) even while the stackable element **732** is stacked on top of the fan assembly **700**. In other words, when the mating interface **724** is formed on the fan housing **704**, the fan housing **704** is not able to move between the stored position and the deployed position while the stackable element **732** is stacked on top of the fan assembly **700**, and when the mating interface **724** is formed on the base **716**, the fan housing **704** is able to move between the stored position and the deployed position while the stackable element **732** is stacked on top of the fan assembly **700** such that the fan housing **704** is operable to generate airflow in a plurality of directions even when the fan assembly **700** and the stackable element **732** are stacked. The mating interface **724** further supports stacking and interlocking of the fan assembly **700** on top of the stackable element **732**. In some embodiments, the fan assembly **700** can be stacked on top of the stackable element **732** (e.g., on a lid of the stackable element **732**) in a side-by-side arrangement with another fan assembly **700** or another stackable element **732**.

(108) As illustrated in FIGS. 23A-23D, the base **716** includes a first wall **716a** having a first part of the mating interface **724** configured to interact with the stackable element **732**, a second wall **716b** opposite the first wall **716a** having a second part of the mating interface **724** configured to interact

with the stackable element **732**, and a wall or a plurality of walls **716c**, extending between the first wall **716a** and the second wall **716b**, defining an opening **736** to the storage area **720** in the base **716**. In the deployed position, the fan housing **704** is at least partially extended out of the storage area **720** through the opening **736**. In the illustrated embodiment, the plurality of walls **716c** defines an outer perimeter of the base **716**. In some embodiments, the outer perimeter is polygonal. In some embodiments, the outer perimeter is circular. In the illustrated embodiment, the outer perimeter is rectangular, and may even be square. As illustrated in FIG. **23C**, the fan assembly **700** can be stacked on the stackable element **732** in an upright position in which the first wall **716a** and the second wall **716b** are positioned at opposing upper and lower sides of the base **716**, and the plurality of walls **716c** are positioned on sides of the base **716**.

(109) Referring now to FIGS. **23C** and **23D**, the fan assembly **700** further includes a joint or coupler **740** moveably connecting the fan housing **704** to the base **716**. The coupler **740** has a first end **740a** pivotally connected to the base **716** about a first axis **744**, and a second end **740b** rotatably connected to the fan housing **704** about a second axis **748** that is transverse relative to the first axis **744**. The fan housing **704** is configured to be selectively pivoted relative to the base **716** about the first axis **744** to move the fan housing **704** into and out of the storage area **720** and selectively rotated relative to the base **716** about the second axis **748** to change a direction of airflow generated by the fan **712**. In other words, the coupler **740** is moveable to alter both a pitch and a yaw of the fan housing **704** relative the base **716**. As best illustrated in FIG. **23C**, both the pitch and the yaw of the fan housing **704** (e.g., first axis **744** and second axis **748**) may be altered while the stackable element **732** is stacked on the fan assembly **700** to move the fan housing **704** between the stored position and the deployed position. As such, the fan housing **704** may move about two degrees of freedom relative to the base **716**.

(110) In some embodiments, as illustrated in FIGS. **22A** and **23B**, a detent mechanism **752** may releasably hold the fan housing **704** in set positions (e.g., 0 degrees, 45 degrees, 90 degrees, and 135 degrees) relative to the base **716**. In other words, the detent mechanism **752** is configured to be set to a predetermined position by a user, where the predetermined position corresponds to an angular position of the fan housing **704** relative to the base **716**. In some embodiments, the fan assembly **700** even further includes a handle **756** configured to accommodate carrying of the fan assembly **700** while stacked with or unstacked from the stackable element **732**. The handle **756** may be positioned on the fan housing **704**, the base **716**, or both.

(111) In some embodiments, as illustrated in FIGS. **24** and **25**, the fan housing **704** may be smaller than the outer perimeter of the fan assembly **700** or base **716**, and the fan assembly **700** may further include a storage compartment **760** separate from the storage area **720**. The storage compartment **760** may hold tools, tool bits, and/or the like. In some embodiments, the battery receptacle **715** and/or AC power inlet **717** are positioned within the storage compartment **760**. The storage compartment **760** may also be waterproof and or sealable.

(112) Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described. Various features and advantages of the invention are set forth in the following claims.

Claims

1. A battery powered box-type fan assembly comprising: a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a battery receptacle positioned on the fan housing, the battery receptacle configured to receive a removable battery pack to power the motor; and a base coupled to the fan housing and configured to be placed on a support surface, the base elevating the fan housing above the support surface and operable to support the fan housing in multiple

configurations, the base including a body, arms extending upwardly from the body and coupled to the fan housing on opposite sides of the airflow chamber, the arms defining a pivot axis for the fan housing, and a stand having a first end pivotally coupled to the body and a second end opposite the first end, the stand being pivotable relative to the fan housing and the body between a collapsed position, in which the stand is flipped up toward the fan housing and the body such that the stand is positioned adjacent a side of the arms and the second end is positioned closer than the first end to the pivot axis, and a deployed position, in which the stand extends away from the fan housing and the body and is configured to contact the support surface; wherein the fan housing is movable relative to the body and the arms about the pivot axis.

2. The battery powered box-type fan assembly of claim 1, wherein the stand includes a pair of legs moveable with the stand.
3. The battery powered box-type fan assembly of claim 2, wherein the pair of legs are attached together by an elongated hinge member.
4. The battery powered box-type fan assembly of claim 1, wherein the stand is a first stand positioned on a first side of the base, and wherein the base further includes a second stand positioned on a second side of the base, the second stand being pivotable relative to the fan housing between a collapsed position, in which the second stand is flipped up toward the fan housing, and a deployed position, in which the second stand extends away from the fan housing and is configured to contact the support surface.
5. The battery powered box-type fan assembly of claim 4, wherein the first stand includes a first pair of legs, wherein the second stand includes a second pair of legs, wherein ends of the first pair of legs and the second pair of legs are rotatable downwardly and away from the fan housing when rotated from the collapsed position to the deployed position.
6. The battery powered box-type fan assembly of claim 5, wherein the first pair of legs and the second pair of legs are lockable into the deployed position.
7. The battery powered box-type fan assembly of claim 4, wherein the first stand and the second stand are rotatable relative to the fan housing independently from one another.
8. The battery powered box-type fan assembly of claim 1, wherein the base includes a latch coupled to the stand, the latch releasably securing a position of the stand, the latch being manually operable to allow movement of the stand relative to the fan housing.
9. The battery powered box-type fan assembly of claim 1, further comprising an AC power inlet, wherein the AC power inlet is electrically coupled to the motor, to the battery receptacle, or to both.
10. The battery powered box-type fan assembly of claim 9, wherein the AC power inlet is configured to provide charging power to the removable battery pack received in the battery receptacle.
11. The battery powered box-type fan assembly of claim 9, wherein the AC power inlet is positioned on the fan housing.
12. The battery powered box-type fan assembly of claim 1, wherein the fan housing further includes a first face adjacent an inlet side of the airflow chamber, and wherein the first face includes a retainer configured to receive and retain a replaceable filter on the first face.
13. The battery powered box-type fan assembly of claim 12, wherein the fan housing further includes a second face opposite the first face and adjacent an outlet side of the airflow chamber, and wherein the battery receptacle is accessible from the second face and located on a lower end of the fan housing.
14. The battery powered box-type fan assembly of claim 1, wherein the battery receptacle is configured to receive the removable battery pack in both the collapsed position and the deployed position.
15. A battery powered box-type fan assembly comprising: a fan housing that at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a battery receptacle positioned on the fan

housing, the battery receptacle configured to receive a removable battery pack to power the motor; and a base coupled to the fan housing and configured to be placed on a support surface, the base including a body elevating the fan housing above the support surface and operable to support the fan housing in multiple configurations, a stand movable relative to the body, and a latch coupled to the body and the stand, the latch releasably securing a position of the stand relative to the body, the latch being manually operable to allow movement of the stand relative to the body; wherein the latch includes a ratchet mechanism having a series of ramps and shoulders configured to contact a complementary series of ramps and shoulders formed on the stand to permit ratcheting rotation of the stand relative to the body in a first direction and inhibit rotation of the stand relative to the body in a second direction.

16. The battery powered box-type fan assembly of claim 15, wherein the stand is movable relative to the body between a collapsed position and a deployed position, wherein the first direction of the ratchet mechanism corresponds with rotation of the stand relative to the body from the collapsed position to the deployed position, and wherein the second direction of the ratchet mechanism corresponds with rotation of the stand relative to the body from the deployed position to the collapsed position.

17. A battery powered box-type fan assembly comprising: a fan housing having a first face, a second face opposite the first face, and a perimeter made of a plurality of faces extending between the first and second faces; the first face, the second face, and the perimeter defining a volume therebetween, and the fan housing at least partially defines an airflow chamber; a plurality of fan blades positioned within the airflow chamber; a motor operable to rotate the plurality of fan blades; a base pivotally coupled to the fan housing and configured to be placed on a support surface, the base including arms elevating the fan housing above the support surface, wherein the fan housing is pivotable relative to the arms about a pivot axis that is parallel to the support surface; a battery receptacle positioned at a first corner formed by two adjacent faces of the perimeter of the fan housing below the pivot axis and electrically coupled to the motor; an AC power inlet positioned at a second corner formed by two adjacent faces of the perimeter of the fan housing below the pivot axis and electrically coupled to the motor; and a battery pack removably coupled to the battery receptacle, the battery pack being electrically coupled to the motor through the battery receptacle, the battery pack positioned within the volume of the fan housing; wherein the battery pack rotates with the fan housing relative to the arms about the pivot axis when positioned within the volume of the fan housing; and wherein the first face of the fan housing includes a retainer configured to selectively receive a replaceable filter and retain the replaceable filter on the fan housing, and wherein the battery pack is accessible on the second face of the fan housing.

18. The battery powered box-type fan assembly of claim 17, wherein the plurality of fan blades rotate about a rotation axis that is perpendicular to the pivot axis, and wherein the battery pack is further positioned beneath the rotation axis of the plurality of fan blades.

19. The battery powered box-type fan assembly of claim 17, wherein the perimeter of the fan housing is polygonal.
